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Stable Spike: Dual Consistency Optimization via Bitwise AND Operations for Spiking Neural Networks

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Abstract

*Although the temporal spike dynamics of spiking neural networks (SNNs) enable low-power temporal pattern capture capabilities, they also incur inherent inconsistencies that severely compromise representation. In this paper, we perform dual consistency optimization via **Stable Spike** to mitigate this problem, thereby improving the recognition performance of SNNs. With the hardware-friendly "AND" bit operation, we efficiently decouple the stable spike skeleton from the multi-timestep spike maps, thereby capturing critical semantics while reducing inconsistencies from variable noise spikes. Enforcing the unstable spike maps to converge to the stable spike skeleton significantly improves the inherent consistency across timesteps. Furthermore, we inject amplitude-aware spike noise into the stable spike skeleton to diversify the representations while preserving consistent semantics. The SNN is encouraged to produce perturbation-consistent predictions, thereby contributing to generalization. Extensive experiments across multiple architectures and datasets validate the effectiveness and versatility of our method. In particular, our method significantly advances neuromorphic object recognition under ultra-low latency, improving accuracy by up to 8.33%. This will help unlock the full power consumption and speed potential of SNNs.*

1. Introduction

The rapid development of deep learning, especially the emergence of large models, makes the power consumption of AI models a non-negligible challenge. The human brain, on the other hand, enjoys a significant power consumption advantage over artificial neural networks (ANNs) [28], making neuromorphic computing a promising low-power, high-performance computing paradigm [23]. Spiking neural networks (SNNs) transmit sparse binary spikes over multiple timesteps, requiring only addition operations when deployed on neuromorphic chips, and consuming significantly less power compared to ANNs [41–43]. Similarly, neuromorphic data represents information as sparse binary events, and recognizing neuromorphic objects using SNNs can be inherently low-power and low-latency [3, 36, 43].

However, although temporal spike dynamics give SNNs low-power spatio-temporal representations, they also introduce the risk of inconsistency. Differences in neuronal states and input currents across timesteps lead to excessive variability in spike maps and predictions, negatively affecting overall performance [4]. To address this problem, [4] indirectly promotes spike consistency through membrane potential smoothing and maintains the overall stability by performing logit distillation with adjacent timesteps. Although effective, this indirect strategy requires the modification of neuronal dynamics and struggles to be readily adopted as a versatile SNN enhancement solution, especially for deployment on neuromorphic chips where neuron models are often predetermined [14, 22, 43]. Therefore, efficiently and versatilely promoting the predictive consistency and performance of SNNs remains an open challenge.

In this paper, we propose the **Stable Spike**, which efficiently improves the consistency and performance of the SNN without any modification to the spiking neurons or architecture. Our method stems from a key insight: *despite differences, a well-optimized SNN can capture object-critical features at different timesteps*. This is illustrated by the spike map visualization in Fig. 1, where spike maps of different timesteps capture object-relevant features, while excessive redundant noise spikes unrelated to the object lead to variability. Therefore, we can decouple and extract the critical information, i.e. the stable spike skeleton, from the variational spike maps. To do this, we perform a hardware-friendly "AND" bit operation [25, 40] on the spike maps of adjacent timesteps to efficiently retrieve consistent and stable 1-value spikes while ignoring messy spikes. As shown in Fig. 1, the stable spike maps reduce variability and outline a clear feature skeleton. We encourage the original spike maps to converge to the stable spike during training, which directly reduces the discrepancy between the multi-timestep spike maps and mitigates the in-

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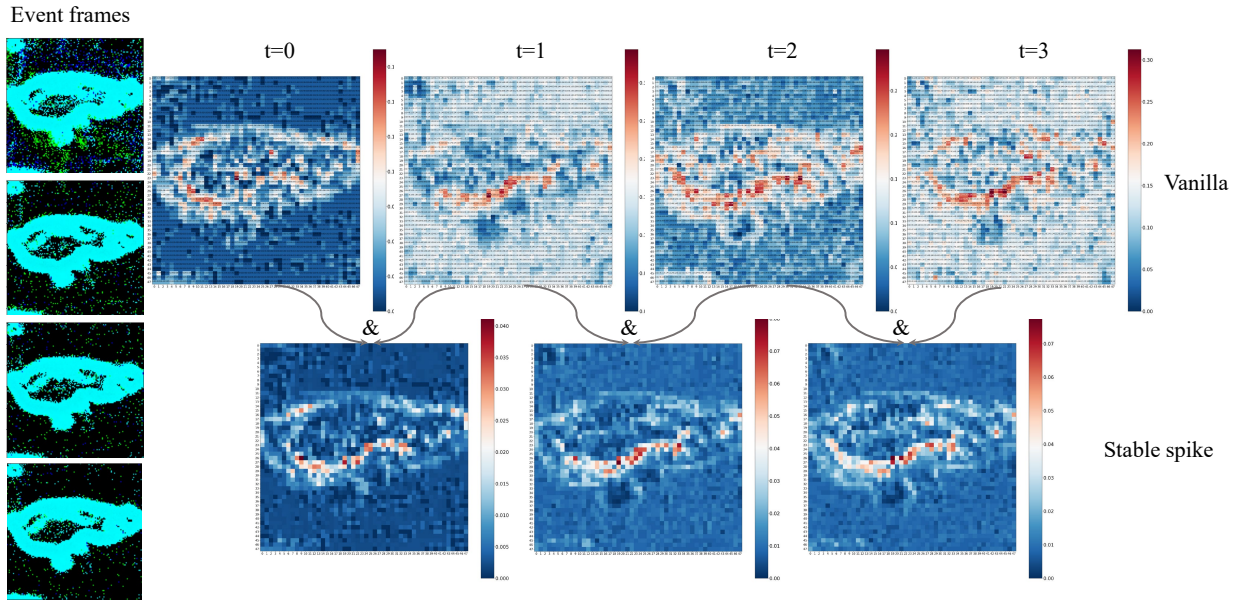


Figure 1. Comparison of vanilla SNN spike maps and stable spike maps. Vanilla spike maps varied widely across timesteps, negatively affecting the overall representation; stable spike maps, decoupled by minimal & operation, consistently represented the feature skeleon. The event frames are from the CIFAR10-DVS dataset and the visualization shows the spike maps of the first layer in VGG-9 after averaging over all channels. Additional visualizations in **Supplementary Material** show that the stable spike consistently extracts the feature skeleton.

terference of redundant, varying noise spikes.

While consistency across timesteps stabilizes the overall prediction, moderate diversity promotes generalization and convergence [9, 19]. However, unlike ANNs that can directly benefit from random Gaussian noise [19], SNNs necessitate that (1) the noise to be discrete or else a mismatch in training-inference precision will occur, and (2) the noise amplitude to be appropriate or else it will interfere with normal training. To this end, we propose amplitude-aware spike noise, which possesses the discrete properties of binary spikes and generates noise depending on the amplitude of the spike firing rate. This enables the noise to adequately contribute to the generalization of high-amplitude elements while preventing low-amplitude elements from being overly perturbed. We inject amplitude-aware spike noise into the stable spike firing rate skeleton to preserve critical object semantics and push the perturbed predictions to approximate the original predictions. This enhances the perturbation consistency of SNNs to unknown variations, thereby improving generalizability.

Facilitated by dual consistency, our method efficiently boosts the performance of the SNN without changing the neurons or the model architecture. This allows our method to be plug-and-play for different spiking neurons and architectures, and to synergize with other methods to enhance performance (as shown in Table 9) with excellent versatility. In particular, our method significantly enhances performance on neuromorphic datasets, boosting the accuracy of DVS-Gesture by up to 8.33% under ultra-low latency at two timesteps. Our contributions are summarized below:

- We propose to decouple the stable spike feature skeleton across timesteps from the SNN by the efficient "AND" operation, and provide consistency guidance to the unstable spike maps.
- We propose to inject amplitude-aware spike noise into the stable spike firing rate to increase feature diversity and improve generalization by enhancing the perturbation prediction consistency of the SNN.
- Extensive experiments across multiple architectures and datasets demonstrate the effectiveness and versatility of our method, especially at enabling neuromorphic object recognition with ultra-low latency.

2. Related Work

Spiking Neural Network. SNNs mimic the information transfer mechanism of the biological nervous system, where binary spike signals are transmitted between spiking neurons over multiple timesteps [28, 36, 42]. Additionally, spiking neuron dynamics can capture implicit temporal dependencies, enabling SNNs to exhibit superior spatio-temporal properties [26]. This allows SNNs to be efficiently deployed on neuromorphic chips, providing performance comparable to that of ANNs while consuming substantially less power [41, 43]. Previous studies have focused on improving the neuron dynamics [7, 15], model architectures [34, 42, 48], and training algorithms [2] to unleash the potential of SNNs. However, the inconsistency of SNNs across timesteps remains an unsolved issue that limits their performance, especially at low latency. Although [4] modifies the neural dynamics and guides the output between

neighboring timesteps to alleviate this problem, further solutions that are more efficient and effective are still needed. To this end, this paper performs dual consistency optimization via the efficient stable spike, without any modification of neurons or architectures, and compatible with other improvements.

Neuromorphic Object Recognition. Unlike traditional static images, neuromorphic sensors such as event cameras represent data as sparse binary events with high temporal resolution and are able to capture motion information about the object [3, 20, 30]. The dimensionality of neuromorphic data can be denoted as $[t, x, y, p]$, where t is the temporal index, $[x, y]$ are the spatial coordinates of the data, and p is the polarity (positive polarity indicates that pixel brightness has increased above a certain threshold, and negative polarity is the opposite). This spatio-temporal property makes neuromorphic objects naturally suited for recognition by SNNs, and the power and speed benefits of combining the two have been widely demonstrated [20, 43]. There has been extensive work on optimizing SNNs for neuromorphic object recognition, but they still suffer from significant latencies [2, 8, 21, 46] (typically more than 10 timesteps). In particular, [4] points out that the temporal instability of SNNs is particularly pronounced in neuromorphic data and is a bottleneck limiting their low-latency performance. For these reasons, this paper focuses on enhancing neuromorphic object recognition with SNNs at ultra-low latencies. But notice that, as demonstrated in the experimental section, our method is also effective for static data.

Consistency in SNNs. Inspired by ANNs, previous methods have improved the spatial feature consistency of SNNs through knowledge distillation [12, 38, 39] and contrastive learning [27, 47]. However, focusing solely on spatial aspects fails to leverage the spatio-temporal properties of SNNs fully, especially hindering their integration with neuromorphic data. To improve temporal consistency, especially in neuromorphic recognition performance, further studies distill the temporal dimension [5, 6, 51] or promote consistency in representations between neuromorphic and static objects [13]. However, while consistency has been facilitated, the lack of a stable anchor of consistency has limited the effectiveness of these methods. In this paper, we decouple the stable spikes that outline the feature skeleton as an anchor through the efficient "AND" operation, thus guiding the variable spike maps towards consistency.

3. Method

We first describe the spiking neuron dynamics and the inconsistency problem in SNNs, and then detail how dual consistency optimization can be efficiently achieved by stable spike to improve the performance of SNNs.

3.1. Inconsistency Arising from Spike Dynamics

The spiking neuron is the core element in SNNs that mimics the biological charge-firing mechanism to generate binary spike signals. In this paper, we use the most commonly used leaky integrate-and-fire (LIF) neuron model [4, 37], which balances biological plausibility with ease of implementation. The LIF neuron model iteratively undergoes three phases of charging, firing, and resetting within $T \geq 1$ discrete timesteps. Upon receiving an input current I from presynaptic neurons, the LIF neuron incorporates it into the membrane potential H, i.e., charging. The LIF neuron then determines whether the membrane potential exceeds a specific firing threshold ϑ to generate a spike S or remain silent. During the resetting phase, the LIF neuron updates the membrane potential based on the fired spike, subtracting the membrane potential by the same amplitude as the threshold in a soft reset manner, and leaving it unchanged if no spike is fired. The iterative dynamics of LIF neurons within a single timestep can be expressed as:

$$H^l_{i,t} = (1 - \frac{1}{\tau}) H^l_{i,t-1} + I^l_{i,t}, \tag{1}$$

$$S^l_{i,t} = \begin{cases} 1, & H^l_{i,t} \geq \vartheta \\ 0, & H^l_{i,t} < \vartheta \end{cases}, \tag{2}$$

$$H^l_{i,t} = H^l_{i,t} - S^l_{i,t}\vartheta, \tag{3}$$

where l, i, and t denote the layer, neuron, and timestep indexes in the SNN, respectively. In particular, the membrane potential of LIF neurons leaks when the dynamics iterate to the next timestep, which is controlled by τ in Eq. (1) with a default value of 2.0.

The output of a spiking neuron at each timestep depends on both the input I and its membrane potential value H, since the membrane potential is maintained across timesteps. The synergistic interaction of these two factors results in differences in the output of the SNN across timesteps. As shown in Fig. 1, excessive differences between spike maps across timesteps lead to overall inconsistency, and plenty of highly variable spurious spikes impeding semantic feature extraction. In particular, since the membrane potential of the spiking neuron is usually initialized to 0, the output of the early timestep is more confusing compared to the late one [4, 37]. The inconsistency across timesteps limits the overall performance of the SNN, especially when only early timesteps are available for low-latency inference.

3.2. Stable Spike for Spike Map Consistency

To promote multi-timestep consistency of SNNs, we separate the variable spike feature maps into consistent stable spike skeletons and unstable spurious redundant spikes. Therefore, we are able to train the spike feature maps to converge to a stable and consistent skeleton while ignoring the interference of redundant and variable spurious spikes.

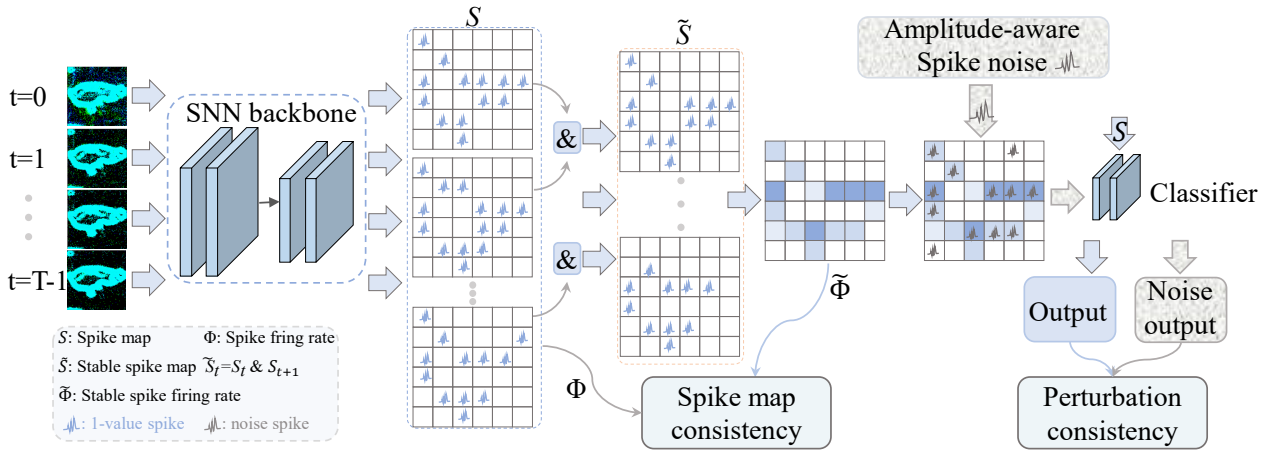


Figure 2. The stable spikes decoupled by the minimal & operation are used as the anchor. On the one hand, we promote the variable original spike maps to converge to the stable spike firing rate skeleton, i.e., spike map consistency. On the other hand, we introduce amplitude-aware spike noise to the stable spiking firing rate to preserve the key semantics and increase the feature diversity, allowing the SNN to be insensitive to the perturbation and promote generalization, i.e., perturbation consistency.

Without loss of generality, we denote the spike feature map generated by the SNN over T timesteps as $S \in \{0,1\}^{T\times B\times C\times H\times W}$, where B is the batch size, C is the number of channels, and H and W denote the height and width, respectively (temporarily ignore layer indices). To efficiently extract salient semantic features captured by the SNN over T timesteps from the unstable S, we use the binary data-friendly "AND" (&) operation to compute the stable spike skeleton between adjacent spike maps. In particular, the & operation is able to preserve 1-valued elements coexisting in the two binary data while eliminating unstable noise elements, and is naturally suited for modeling consistency in SNNs. Among the T varying spike maps, we extract a total of $T-1$ stable spikes, which is expressed in mathematical form as follows:

$$\tilde{S}_{i,t} = S_{i,t} \,\&\, S_{i,t+1} = \begin{cases} 1, & S_{i,t} = S_{i,t+1} = 1 \\ 0, & else \end{cases}, \quad (4)$$

Fig. 1 visualizes the stable spikes decoupled by the & operation, which reveal a clear feature skeleton with significantly less inconsistency compared to the original spike maps.

Furthermore, we define the stable spike firing rate, $\tilde{\Phi} = \frac{1}{T-1}\sum_{t=0}^{T-2}\tilde{S}_t$, to represent the aggregated feature of stable spikes over multiple timesteps. $\tilde{\Phi}$ effectively represents the feature skeleton extracted by the SNN over multiple timesteps, so we align the variable spike map S to it to improve consistency and ignore spurious spike interference. To this end, we take the stable spike firing rate $\tilde{\Phi}$ as an anchor point to guide the spike map S_t, and achieve the consistency constraint by backpropagation of the spike map consistency objective function.

During implementation, we use the mean squared error (MSE) function as the consistency guidance function and perform averaging on the original T spike feature maps to obtain the spike fring rate Φ. Thus, the mathematical form of spike map consistency alignment can be expressed as:

$$\begin{aligned} \mathcal{L}_{spike} &= MSE(\tilde{\Phi}, \Phi) \\ &= \frac{\sum_{i=1}^{C\times H\times W}(\frac{1}{T-1}\sum_{t=0}^{T-2}\tilde{S}_{i,t} - \frac{1}{T}\sum_{t=0}^{T-1}S_{i,t})^2}{C\times H\times W} \end{aligned} \quad (5)$$

Table 1. Alternative consistency function.

Dataset	Vanilla	$\mathcal{L}_{spike}$			$\mathcal{L}_{noise}$		
		MSE	KL	Cosine	MSE	KL	Cosine
CIFAR10-DVS	72.9	**77.1**	76.2	75.6	76.1	**77.1**	76.0
DVS-Gesture	87.15	**94.44**	94.10	93.40	93.40	**94.44**	91.67

During training, the gradient of $\mathcal{L}_{spike}$ directly contributes to the stabilization of the spike firing rate, while preventing performance degradation due to excessive variance over multiple timesteps. Compared to the indirect adjustment of membrane potential and final output [4], the direct control of spike firing rate brings more significant consistency and superior discrimination performance. It is worth noting that our method is not limited to a specific consistency function. The results in Table 1 show that significant performance gains can also be achieved with KL divergence or cosine similarity, suggesting that our method promises to further unlock the performance of SNNs with an optimized consistency function implementation.

3.3. Amplitude-Aware Spike Noise for Perturbation Consistency

The feature diversity of neural networks can facilitate generalization under the premise of consistently preserving semantic information [9, 19]. Inspired by this, we expect to further increase the spike feature diversity to promote the generalization of SNNs. However, while ANNs can benefit from vanilla Gaussian noise [19], this does not work for SNNs due to the following two spike-specific character-

istics: (1) the inherent discrete property of binary spikes makes the addition of continuous Gaussian noise during training lead to training-inference precision mismatches; and (2) discrete spike firing rate is more sensitive to noise amplitude compared to floating-point numerical activation. Therefore, we propose amplitude-aware spike noise, which ensures that the SNN conveys discrete information through noise spikes during both training and inference, while the noise amplitude depends on the spike firing rate, ensuring that the perturbation promotes generalization without causing degradation.

A prerequisite for noise perturbation of features to facilitate generalization is the preservation of key semantic information [35]. Since the stable spike firing rate $\tilde{\Phi}$ filters out differences across timesteps as a stable feature skeleton, we add noise to $\tilde{\Phi}$ instead of directly perturbing the original features as in an ANN [19, 35]. The discrete value range of the stable spike firing rate $\tilde{\Phi}$ decoupled within T timesteps is $\{0, \frac{1}{T-1}, \cdots, \frac{T-2}{T-1}, 1\}$. To maintain the discrete property, we sample binary spike noise $\varepsilon \in \{0, 1\}$ from the Bernoulli distribution with probability p and add it to the stable spike firing rate $\tilde{\Phi}$ for perturbation. Specifically, the spike noise ε is obtained by randomly sampling the value p_{tmp} from $[0, 1]$ and comparing it to the noise probability p. If p_{tmp} is less than or equal to p, a spike is generated. This is similar to value-dependent rate encoding of inputs, but requires only one random number generation and comparison, whereas input encoding requires multiple random samples to approximate the Poisson distribution [10, 17, 29]. After perturbation by the noise spike, the value range of the perturbed firing rate is $\{0, \frac{1}{T-1}, \cdots, \frac{T-2}{T-1}, 1, 1+\frac{1}{T-1}, \cdots, 1+\frac{T-2}{T-1}, 2\}$, which remains the discrete property.

On the one hand, if the spike feature value is small, excessive noise will disrupt the key semantics and cause performance degradation; on the other hand, a large spike feature value is insensitive to small noise perturbations, making it difficult to promote generalizability. To this end, we generate amplitude-aware spike noise by applying small perturbations to low firing rate elements and significant perturbations to higher firing rate elements, with the stable firing rate $\tilde{\Phi}$ as the reference. For the implementation, we take the firing rate value of the corresponding position element in $\tilde{\Phi}$ as the spike noise probability, i.e., $p_{c,i,j} = \tilde{\Phi}_{c,i,j}$, where (c, i, j) denotes the position index. The amplitude-aware pulse noise can be expressed as:

$$\varepsilon_{c,i,j} = \begin{cases} 1, & \text{with probability } \tilde{\Phi}_{c,i,j} \\ 0, & else \end{cases}. \tag{6}$$

In this way, elements with a higher stable spike firing rate are more likely to be perturbed by a 1-value spike, and vice versa, elements with low firing rate are prevented from being excessively perturbed.

Algorithm 1 The stable spike algorithm.

Input: training dataset $\{x_i, y_i\}_{i=1}^{B}$, timestep T
Initialization of parameters θ in the SNN $f(\cdot)$
for $t = 0$ **to** $T - 1$ **do**
  Forward propagation calculates the backbone spike map $S_t = f(x)$
  if $t > 0$ **then**
    Calculate the stable spike $\tilde{S}_{t-1} = S_{t-1} \& S_t$
  end if
  Forward propagation produces the output O_t
end for
Adding amplitude-aware spike noise ε to the stable spike firing rate $\Phi_{noise} = \tilde{\Phi} + \varepsilon$
Calculate the noise output O_{noise}
Calculate the dual consistency losses ← Eq. 9
Backpropagation to optimize parameters θ
Output: Consistency-optimized SNN

After performing the noise perturbation on the stable spike firing rate to obtain $\Phi_{noise} = \tilde{\Phi} + \varepsilon$, we forward propagate the perturbed Φ_{noise} to generate the final prediction O_{noise}. To allow the SNN to learn generalized representations, we align the probability distributions of the noisy prediction O_{noise} with the original clean prediction O. The temperature α softening output logit is first used to obtain its probability distribution:

$$p_j = \frac{e^{O_j/\alpha}}{\sum_{k=1}^{K} e^{O_k/\alpha}}, p_{noise,j} = \frac{e^{O_{noise,j}/\alpha}}{\sum_{k=1}^{K} e^{O_{noise,k}/\alpha}}, \tag{7}$$

where j denotes the j-th class, K denotes a total of K classes, and temperature α is set to 2. We use the KL divergence to promote consistency of the probability distribution after softening:

$$\mathcal{L}_{noise} = \alpha^2 KL(O||O_{noise}) = \alpha^2 \sum_{k=1}^{K} O_k log(\frac{O_k}{O_{noise,k}}). \tag{8}$$

It is worth noting that the original SNN output O averaged over T timesteps is used as the final output; the noise prediction without temporal dimension is used directly as the final output. In addition, we compute stable spikes and perform consistency guidance only for the backbone features of the SNN, allowing additional forward propagation only past the classifier with negligible overhead.

During training, the spike map consistency loss (Eq. 5) and perturbation consistency loss (Eq. 8) are seamlessly combined with the cross-entropy loss of the classification task by balancing coefficients β and γ:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \beta \mathcal{L}_{spike} + \gamma \mathcal{L}_{noise}. \tag{9}$$

The complete algorithm for training SNNs by stable spike is shown in Algorithm 1.

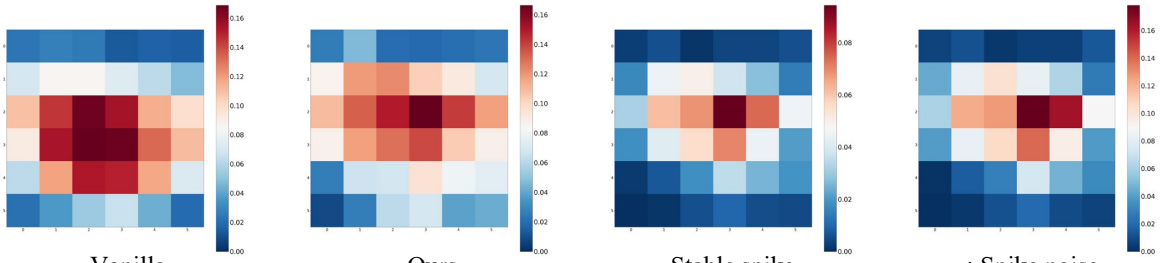


Figure 3. Comparison of spike firing rates. Our spike firing rate shows a clearer feature profile compared to the vanilla SNN by converging to the stable spike and reducing spike interference. Additionally, the proposed amplitude-aware spike noise preserves the key semantic features of stable spikes while increasing feature diversity, thereby improving generalization. To reflect the implementation, we visualize the SNN backbone output of VGG-9 with a spike map size of 6×6.

Table 2. Ablation results across different architectures.

Dataset	Method	VGG-9	ResNet-18	QKFormer
CIFAR10-DVS	Baseline	72.9	66.1	81.2
	$+\mathcal{L}_{spike}$	$75.2_{+2.4}$	$69.7_{+3.6}$	$82.5_{+1.3}$
	$+\mathcal{L}_{noise}$	$75.4_{+2.6}$	$68.5_{+2.4}$	$82.0_{+0.8}$
	+Both	$\mathbf{77.1}_{+4.2}$	$\mathbf{70.3}_{+4.2}$	$\mathbf{82.9}_{+1.7}$
DVS-Gesture	Baseline	87.15	81.59	93.75
	$+\mathcal{L}_{spike}$	$91.32_{+4.17}$	$84.38_{+2.79}$	$94.44_{+0.69}$
	$+\mathcal{L}_{noise}$	$94.09_{+6.94}$	$83.68_{+2.09}$	$94.79_{+1.04}$
	+Both	$\mathbf{94.44}_{+7.29}$	$\mathbf{85.42}_{+3.83}$	$\mathbf{95.49}_{+1.74}$

4. Experiments

We focus on neuromorphic benchmarks CIFAR10-DVS [18], DVS-Gesture [1], and N-Caltech101 [24], while also conducting validation on static CIFAR10/100 and ImageNet using VGG, ResNet, and Transformer-style architectures to demonstrate the effectiveness and generalizability of our method. For VGG-9 and ResNet-18, we use the same training strategy as MPS [4]; for QKFormer, we also follow its original strategy [48]. If not specified, the consistency loss balance coefficients β and γ are set to 1.0, and the timestep is 4 to reflect the low-latency performance of the SNN. The detailed experimental setup can be found in the **Supplementary Material**.

4.1. Ablation Study

To demonstrate the effectiveness and versatility of the proposed method, we performed ablation studies with three architectures on CIFAR10-DVS and DVS-Gesture, and the results are shown in Table 2. The results show that the proposed method consistently delivers performance gains for different SNN architectures, in particular up to 7.29% for DVS-Gesture recognition with VGG-9. When applied to the QKFormer [48] with SOTA performance, the performance is further improved, demonstrating the superior generalizability of our method.

In addition, we examine the effects of two key components of amplitude-aware spike noise: amplitude adaptation and spike noise. We perturbed the stable spike firing rate with a fixed noise spike probability p and continuous Gaussian noise with mean value 0, respectively, and the results are shown in Table 3. The results show that using the fixed noise spike probability whose amplitude is uncontrollable significantly degrades the performance of the proposed method; replacing the discrete spike noise with continuous Gaussian noise also leads to degraded performance. In particular, when the fixed noise probability is excessively high, leading to overly strong perturbations ($p = 0.75$), the model fails to converge. In contrast, the proposed amplitude-aware spike noise considers the discrete, noise-sensitive properties of SNNs and demonstrates superior performance.

Table 3. The effectiveness of amplitude adaption and discrete spike noise. p is the fixed noise spike probability and std is the standard deviation of the continuous Gaussian noise.

Dataset	Ablation	Acc. (%)	Dataset	Ablation	Acc. (%)
CIFAR10-DVS	$p = 0.4$	74.8	DVS-Gesture	$p = 0.4$	87.15
	$p = 0.5$	75.1		$p = 0.5$	88.89
	$p = 0.6$	75.5		$p = 0.6$	86.81
	$std = 0.1$	74.9		$std = 0.1$	88.19
	$std = 0.5$	75.3		$std = 0.5$	91.67
	$std = 1.0$	74.5		$std = 1.0$	89.93
	Ours	**77.1**		Ours	**94.44**

Table 4. Comparison (%) of AND, OR, and XOR bit operations.

Dataset	AND	OR	XOR
CIFAR10-DVS	**77.1**	68.9	74.5
DVS-Gesture	**94.44**	88.54	89.58

Table 4 compares the performance of AND and other bit operations (OR and XOR). The AND operation consistently retrieves (1, 1) spike pairs, whereas the OR and XOR operations retrieve the $\{(1, 0), (0, 1), (1, 1)\}$ and $\{(0, 1), (1, 0)\}$ patterns, respectively. The results show that the AND operation achieves optimal performance, which aligns with our insights on consistency. Notably, the OR operation caused significant degradation due to the simultaneous retrieval of consistent/inconsistent spike patterns triggering confusion.

To investigate the influence of β and γ on the performance, we conducted experiments on DVS-Gesture in the range $\{0.1, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0\}$. The results in Fig. 4 show that these coefficients affect the per-

Table 5. Comparative results (%) on neuromorphic datasets. † denotes knowledge transfer from static data.

	Method	Architecture	$T\downarrow$	Acc. (%) $\uparrow$
CIFAR10-DVS	STAA-SNN$^{CVPR'25}$ [46]	VGG-13	16	82.1
	SLT$^{AAAI'24}$ [2]	VGGSNN	10	81.46
	DeepTAGE$^{ICLR'25}$ [21]	VGG-11	10	81.23
	EnOF-SNN$^{NeurIPS'24}$ [12]	ResNet20	10	80.5
	BKDSNN$^{ECCV'24}$ [39]	Wide-7B-Net	8	72.2
	SSNN$^{AAAI'24}$ [3]	VGG-9	5	73.63
	MPS$^{ICLR'25}$ [4]	VGG-9	5	76.77
		VGGSNN	4	83.2
	Ours	VGG-9	4	**77.1**
		VGGSNN	4	**83.7**
	SNN-ViT$^{ICLR'25}$ [34]	SNN-ViT	16	82.3
	SWformer$^{ECCV'24}$ [8]	SWformer	10	82.9
	SEMM$^{NeurIPS'24}$ [49]	Spikingformer	10	80.7
	SpikingResformer$^{CVPR'24}$ [33]	SpikingResformer	10	81.5
	MPS$^{ICLR'25}$ [4]	SpikingResformer	5	80.6
	QKFormer$^{NeurIPS'24}$ [48]	QKFormer	4	81.2
	Ours	QKFormer	4	**82.9**
DVS-Gesture	MPS$^{ICLR'25}$ [4]	VGG-9	5	93.23
	SSNN$^{AAAI'24}$ [3]	VGG-9	5	90.74
	CLIF$^{ICML'24}$ [15]	VGG-9	4	89.58
	TAB$^{ICLR'24}$ [16]	VGG-9	4	87.50
	SLT$^{AAAI'24}$ [2]	VGG-9	4	88.19
	Ours	VGG-9	4	**94.44**
	QKFormer$^{NeurIPS'24}$ [48]	QKFormer	16	98.60
	SpikingResformer$^{CVPR'24}$ [33]	SpikingResformer	16	98.60
	SEMM$^{NeurIPS'24}$ [49]	Spikingformer	10	96.88
	MPS$^{ICLR'25}$ [4]	SpikingResformer	5	94.44
	QKFormer$^{NeurIPS'24}$ [48]	QKFormer	4	93.75
	Ours	QKFormer	4	**95.49**
			16	**98.61**
N-Caltech101	TCJA-TET-SNN$^{TNNLS'24}$ [50]	CombinedSNN	14	82.50
	SWformer$^{ECCV'24}$ [8]	SWformer	10	88.45
	IMP+TET-S$^{NeurIPS'24}$ [31]	VGGSNN	10	85.01
	TKS$^{IEEE\ TAI'24}$ [6]	VGGSNN	10	84.10
	EventMix$^{Inf.Sci.'23}$ [30]	ResNet-18	10	79.47
	TIM$^{IJCAI'24}$ [32]	Spikformer	10	79.00
	NDA$^{ECCV'22}$ [20]	VGG-11	10	78.20
	Knowledge-Transfer$^{AAAI'24}$ [13]	VGGSNN	10	93.18†
	SSNN$^{AAAI'24}$ [3]	VGG-9	5	77.97
	MPS$^{ICLR'25}$ [4]	VGG-9	5	82.71
		VGGSNN	10	93.68†
	Ours	VGG-9	4	**83.92**
		VGGSNN	10	**94.25**†

formance to a certain extent, where the lowest accuracy is 92.01%, while the highest accuracy reaches 95.14%. It is worth noting that although the performance fluctuates, the lower bound of our method still far exceeds the performance of the vanilla SNN (87.15%). This suggests that our method can be plug-and-play to improve performance and offers higher potential after adjusting the balancing coefficients.

4.2. Comparison with Other Methods

Neuromorphic Benchmark. Table 5 shows the comparative results on neuromorphic benchmarks. Without any data augmentation, our VGG-9 achieved a competitive accuracy of 77.1% on CIFAR10-DVS with only 4 timesteps. Using

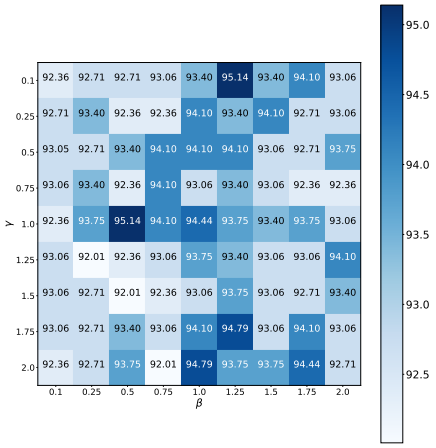


Figure 4. Performance (%) of different balance coefficients β and γ on DVS-Gesture.

Table 6. Comparative results with other methods on ImageNet.

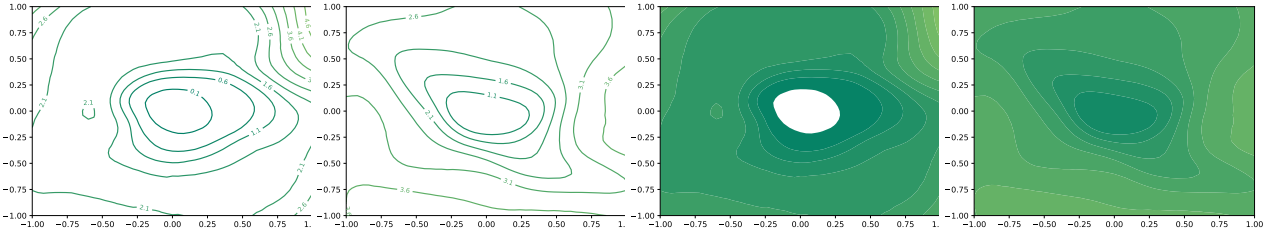
Method	Architecture	T	Acc.(%)
RateBP [44]$^{NeurIPS'24}$	ResNet-34	4	70.01
TAB [16]$^{ICLR'24}$	ResNet34	4	67.78
Shortcut [11]$^{NeurIPS'24}$	ResNet34	4	68.14
FSTA-SNN [45]$^{AAAI'25}$	ResNet34	4	70.23
STAA-SNN [46]$^{CVPR'25}$	ResNet34	4	70.40
IMP+LTS [31]$^{ICLR'25}$	ResNet34	4	68.90
SSCL [47]$^{AAAI'24}$	ResNet34	4	66.78
EnOF [12]$^{NeurIPS'24}$	ResNet34	4	67.40
MPS [4]$^{ICLR'25}$	ResNet-34	4	69.03
Strong2Weak [5]$^{NeurIPS'25}$	ResNet-34	4	70.53
Weak2Strong [5]$^{NeurIPS'25}$	ResNet-34	4	69.87
Ours	ResNet-34	4	**70.59**

standard data augmentation and the VGGSNN architecture, recognition accuracy reaches 83.7%. On DVS-Gesture and N-Caltech101, our method also achieves significant performance advantages at low latency. In particular, our method can be used in conjunction with Knowledge-Transfer [13] to achieve 94.25% accuracy on N-Caltech101 at 10 timesteps, exceeding all other methods.

Static Dataset. Table 6 presents the comparative results on ImageNet. We achieved an accuracy of 70.59% using ResNet-34 with four timesteps, surpassing other comparative methods. On CIFAR10 and CIFAR100, we achieved accuracies of 96.73% and 82.29%, respectively, once again outperforming other methods. See **Supplementary Material** for details.

4.3. Loss Landscape Visualization

To demonstrate the effectiveness of the proposed method from an optimization standpoint, Fig. 5 visually compares the loss landscapes of the vanilla SNN and the proposed method. In Fig. 5(a)(c), the loss distribution of the vanilla SNN is dispersed and irregular, especially the upper-right and the center regions differ significantly. Additionally, local minima interfere with optimization, causing the model



(a) Vanilla (b) Ours (c) Vanilla (d) Ours

Figure 5. Visualization of loss landscapes. (a) (c) The loss landscape of the vanilla SNN exhibits multiple local minima and saddle points. This complexity makes optimization susceptible to falling into local optima. (b) (d) Our method has a smoother and more centralized loss landscape with a clear global minimum, which makes optimization stable and convergent.

Table 7. Scalable performance across timesteps. Our method improves performance consistently from ultra-low to high latencies.

Dataset	$T=2$		$T=4$		$T=6$		$T=8$		$T=10$	
	Vanilla	Ours	Vanilla	Ours	Vanilla	Ours	Vanilla	Ours	Vanilla	Ours
CIFAR10-DVS	72.6	**74.6**$_{+2.0}$	72.9	**77.1**$_{+4.2}$	73.7	**77.5**$_{+3.8}$	75.2	**77.8**$_{+2.6}$	75.5	**78.2**$_{+2.7}$
DVS-Gesture	83.68	**92.01**$_{+8.33}$	87.15	**94.44**$_{+7.29}$	88.19	**94.10**$_{+5.91}$	90.97	**95.49**$_{+4.52}$	92.01	**95.49**$_{+3.48}$

Table 8. Comparison of spike firing rate (%) and power consumption ($\times 10^6 pJ$) on CIFAR10-DVS with VGG-9.

Layer $\rightarrow$	1	2	3	4	5	6	7	8	Power $\downarrow$
Vanilla	**9.24**	4.26	3.38	3.12	2.17	1.45	1.20	7.63	189.83
Ours	10.14	**3.61**	**3.20**	**2.88**	**1.70**	**1.25**	**1.10**	**9.81**	**181.02**

to be highly susceptible to falling into local optimal solutions. In contrast, the loss landscape of our method (Fig. 5(b)(d)) is much smoother overall, without sharp variance despite the presence of spike noise perturbation during training. Consequently, our method is capable of converging to the global optimal solution with enhanced stability and efficiency, thereby facilitating generalization.

4.4. Scalability with Timesteps

Table 7 shows the performance variation of the proposed method with different timesteps. We vary the timestep from the commonly used high latency ($T = 10$) to the ultra-low latency ($T = 2$). The results show that our method yields consistent performance gains that generalize to a wide range of latencies. In particular, the accuracy on DVS-Gesture is improved by 8.33% at $T = 2$, which significantly contributes to the development of ultra-low latency SNNs.

4.5. Spike Firing Rate and Power Consumption

To investigate the influence of our method on spike firing rate and power consumption, we present the comparative results with the vanilla SNN in Table 8. The results show that our method achieves lower firing rates and lower overall power consumption across all layers except the first. This indicates that our method not only improves the performance of SNNs, but also slightly reduces their power consumption during deployment.

4.6. Compatibility with Other Methods

Our method is a plug-and-play SNN enhancement algorithm that requires no modification to the neuron model or architecture. To further validate the compatibility, we combine our method with other SNN methods, including the improved neuron model CLIF [15], the SNN-oriented BN layer TAB [16], and the efficient training algorithm SLT [2]. The results in Table 9 show that our method consistently improves the performance of these methods, suggesting that our method has the potential to efficiently facilitate the performance of more SNN methods in the future.

Table 9. Our method can be seamlessly integrated with other methods and exhibits superior generalizability.

Dataset	Type	Method	Acc. (%)	+ Ours
CIFAR10-DVS	Spiking neuron	CLIF [15]	74.3	**76.2**$_{+1.9}$
	BN Layer	TAB [16]	73.1	**75.4**$_{+2.3}$
	Training	SLT [2]	74.1	**75.9**$_{+1.8}$
DVS-Gesture	Spiking neuron	CLIF [15]	89.58	**95.83**$_{+6.25}$
	BN Layer	TAB [16]	87.50	**92.36**$_{+4.86}$
	Training	SLT [2]	88.19	**90.97**$_{+2.78}$

5. Conclusion

In this paper, we decouple the stable spike skeleton via the "AND" bit operation, and motivate the spike maps to converge towards it for consistency. Furthermore, we inject amplitude-aware spike noise to facilitate spike feature diversity while preserving key semantics, encouraging the SNN to generate perturbation-consistent predictions for generalization. Our method is neuron-architecture agnostic and can therefore be used to boost the performance of other SNN methods as a plug-and-play solution. Extensive experiments demonstrate the effectiveness and performance advantages of our method. We expect this work to inspire advancements in ultra-low-latency, high-performance SNNs.

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